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ORIGINAL ARTICLE

Differential biomarker profiles between unprovoked venous thromboembolism and cancer

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ABSTRACT

Background: The relationship between cancer and venous thromboembolic disease (VTD) are complex because the activated coagulation factors are not only involved in thrombosis but also in malignant processes, such as angiogenesis and metastasis.
Objective: To compare phenotypes of extracellular vesicles (EVs), and levels of D-dimer, soluble P-selectin (sP-selectin) and antigenic tissue factor (TF) between unprovoked VTD patients, who did not develop cancer during one-year follow-up, and those with advanced stage of cancer but not associated with VTD.
Methods: A prospective study in which we included 138 unprovoked VTD patients and 67 advanced cancer patients, who did not develop thrombosis. Levels of EVs of different cellular origin (platelet, endothelium and leukocyte), EVs positive for tissue factor (TF) and P-selectin glycoprotein ligand 1 were quantified by flow cytometry. D-dimer, soluble P-selectin (sP-selectin) and antigenic TF were determined by ELISA.
Results: TF-positive EVs, D-dimer, and sP-selectin were markedly elevated in unprovoked VTD patients compared to cancer patients without association with thrombosis.
Conclusions: Levels of TF-positive EVs, D-dimer and sP-selectin are able to discriminate between unprovoked VTD patients not related to cancer and cancer patients not associated with VTD. These results could lead to the application of EVs as biomarkers of both diseases.
Key messages: Circulating EVs, specifically TF-positive EVs, in combination with plasmatic markers of hypercoagulable states, such as D-dimer, sP-selectin and antigen TF, are able to discriminate between cancer patients without thrombosis and patients with unprovoked VTD. Research fields could be opened. Future studies will assess if these biomarkers together serve as predicting thrombotic events in cancer populations
Abbreviations: BMI: body mass index; CT: computerized tomography; DVT: deep vein thrombosis; EVs: extracellular vesicles; EEVs: endothelial-derived extracellular vesicles; FITC: fluorescein isothiocyanate; LEVs: leukocyte-derived extracellular vesicles; PhE: phycoerythrin; PE: pulmonary embolism; PEVs: platelet-derived extracellular vesicles; PPP: platelet poor plasma; PSGL1 + EVs: extracellular vesicles carrying P-selectin glycoprotein ligand 1; sP-selectin: soluble P-selectin; TF: tissue factor; TF + EVs: extracellular vesicles carrying tissue factor; VTD: venous thromboembolic disease

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Venous thromboembolism; cancer; pulmonary embolism; cellular extravesicles; D-dimer; P-selectin

Introduction

Cancer may be considered as a hypercoagulable state in which changes in the blood coagulation system favours pro-coagulation. A bidirectional relationship exists between cancer and venous thromboembolic disease (VTD) including deep vein thrombosis (DVT) and pulmonary embolism (PE). Approximately 5–15% of cancer patients develop thrombotic events [1,2]. This frequency is increasing due to the use of new prothrombotic anticancer agents and the improvement in detecting incidental VTD with the use of sensitive imaging techniques. Thrombosis is considered a leading cause of death in cancer patients [3,4]. Moreover, certain types of malignancy including

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Supplemental data for this article can be accessed here.

pancreatic and lung cancer and the advanced stage of cancer are associated with an increased risk of thrombosis [5,6]. On the other hand, VTD has been implicated as the earliest sign of cancer, as up to 10% of patients with unprovoked VTD are diagnosed with cancer within one year of the first appearance of thrombotic symptom [7].

Extracellular vesicles (EVs) are phospholipid vesicles from 0.1 μm to 1 μm in diameter and are derived from cells in response to injury, activation or apoptosis. EVs are shed from different cellular types, which determine their biological functions and properties, and have been shown to reflect cellular activation and/or tissue degeneration *in vivo* [8,9]. Moreover, the clinical application of EVs as relevant biomarkers of diagnosis or prognosis of pathophysiological processes has also been considered [10,11]. High levels of total EVs are associated with a variety of diseases related to vascular damage, inflammation, blood coagulation and angiogenesis. Subpopulations of cell-specific EVs have also been found to be elevated in diseases, such as endothelial-derived EVs (EEVs) in ischaemic stroke [8], acute coronary syndrome [12] and diabetes mellitus [13], platelet-derived EVs (PEVs) in severe hypertension and patients with sepsis [14,15] and leukocyte-derived EVs (LEVs) in psoriasis [16].

Elevated levels of procoagulant EVs have been found in patients with cancer [17] and VTD [18,19]. In particular, EVs carrying tissue factor (TF^{+}EVs) and EVs carrying P-selectin glycoprotein ligand 1 ($PSGL1^{+}$EVs) have been related to thrombus formation *in vivo* [20,21], leading to increased interest in the use of EVs as potential biomarkers of VTD and cancer. Most studies have been designed to evaluate the use of EVs as predictors of VTD in cancer patients, albeit with inconclusive results [22,23].

On the other hand, during the past several years, elevated biomarkers of hypercoagulability, such as D-dimer, P-selectin and TF antigen, have been associated with a higher risk of VTD, therefore improving established clinical predictions.

The relationship between cancer and VTD is multifaceted and no validated biomarker is available to identify cancer patients with increased risk of VTD. Part of this complexity is because some activated coagulation factors are not only involved in the development of thrombosis but also may contribute to malignant processes, such as angiogenesis and metastasis. The aim of the present study was to compare subpopulations of EVs and other classical coagulation-related blood parameters, such as D-dimer, soluble P-selectin (sP-selectin) and antigenic tissue factor (TF), in cancer and VTD that could allow us to differentiate these diseases.

Materials and methods

Study design

We designed a parallel observational and prospective study of VTD and cancer. The study was approved by the Ethics Committee of Virgen del Rocío Hospital in Seville, Spain. Written informed consent was obtained from all patients participated in the study, which was carried out in adherence to the principles of the Declaration of Helsinki.

Setting

Two groups of patients were recruited in the VTD and Oncology Units of the Virgen del Rocío Hospital in Seville, Spain from March 2011 to November 2013. One group consisted of unprovoked VTD patients and the second group included stage III B or IV patients with advanced lung, upper intestinal (gastric or pancreas) or colon cancer. To avoid possible confounding effects or mismatched EVs determinations in both patient groups, we only studied unprovoked and non-cancer related VTD patients, and non-VTD associated cancer patients. Additionally, a group of healthy subjects was recruited to determine reference values of EVs and coagulation-related blood parameters. Blood samples were collected at the time of patient recruitment.

Participants

Unprovoked VTD group: inclusion and exclusion criteria

Patients, who were diagnosed with the first confirmed episode of acute unprovoked VTD (DVT or PE) and had neither developed nor were suspected of cancer during a clinical follow-up for 12 months from the onset of thrombotic episode, were included for the current study. Blood samples were collected from all patients on the same day of VTD diagnosis.

Patients were not included if anticoagulation treatment was started due to clinical suspicion more than 72 h prior to VTD diagnosis. Other causes of exclusion were: history of neoplasia, or potential risk factors of thrombosis (surgery, prolonged immobilization for medical prescription, trauma, or hormonal treatment). VTD patients diagnosed or under suspicion of cancer within the year of follow-up were excluded from the study.

Non-VTD associated cancer group: Inclusion and exclusion criteria

Patients, who were diagnosed with stage III B or IV lung, upper intestinal or colon cancer, without symptoms or signs of VTD or incidental VTD as determined by a computerized tomography (CT), and did not present any thrombotic episode during a clinical follow-up for 12 months from the diagnosis of cancer, were included in the current study. Cancer patients with a previous history of VTD or under anticoagulant treatment were not included. Cancer patients diagnosed or under suspicion of VTD within the year of follow-up were excluded from the study. Thromboembolic events were confirmed following the current practice guidelines [24]. In addition, patients with suspicion of thromboembolic events, such as unknown cause of dyspnoea, tachyarrhythmia, arterial hypotension or sudden death without known cause, were also excluded.

Healthy group: inclusion and exclusion criteria

Volunteer subjects were included in the healthy group, who neither had personal history of cancer nor thrombosis. In addition, they were not under anticoagulant treatment, had not undergone surgery during the last 3 months before the recruitment to the study and did not have prolonged immobilizations.

Variables: Demographic characteristics and comorbidities were collected from all subjects included in the study. VTD subtypes (DVT and/or PE) were registered in the VTD group, and the subtype, stage and tumour treatment were collected in the cancer group. During the 12 months of follow-up, outcome variables including all-cause mortality, diagnosis of cancer in the VTD group and diagnosis of VTD in the cancer group were collected. Acute symptomatic VTD was confirmed by ultrasonography for suspected DVT and multidetector CT-scan for suspected PE and histological confirmation was required for the diagnosis of cancer. EVs subpopulations and other blood parameters such as D-dimer, sP-selectin and antigenic TF were determined in plasma samples from the final study population.

Platelet poor plasma preparation

Venous blood was collected with a 21-gauge needle (discarding the first 3 mL) and placed in 3.5 mL of 9NC coagulation 3.2% sodium citrate Vacuette® tubes (Greiner bio-one, North Carolina, USA). The tubes were maintained in a vertical position without agitation and the blood samples were processed within 2 h of extraction. Platelet poor plasma (PPP) samples were obtained by centrifugation as described previously [25]. Briefly, the samples were centrifuged at 1500 g for 30 min at 4 °C. The collection of PPP (supernatant) was stopped 1 cm above the buffy coat to avoid cell contamination. The PPP samples were gently mixed, aliquoted, and stored at −80 °C for later MPs measurement.

EVs measurement by flow cytometry

EVs were characterized in a BD LSR Fortessa flow cytometer (Becton Dickinson, San José, CA, USA). Briefly, 30 μL of frozen PPP were thawed at room temperature and incubated for 30 min at 20–25 °C with specific monoclonal antibodies: mouse anti-human CD45-fluorescein isothiocyanate (FITC), mouse anti-human CD31-phycoerythrin (PhE), mouse anti-human CD41-phycoerythrin–cyanine7 (all from Beckman Coulter, Marseille, France), mouse anti-human CD142-FITC (American Diagnostica, Stamford, CT, USA) and mouse anti-human CD162-PE (Biolegend, San Diego, CA, USA). In parallel, samples were also incubated with pertinent matched isotypes that were used as a negative control. Then 2 μL of annexin V–CF™ Blue (Immunostep, Salamanca, Spain) and 20 μL of annexin V binding buffer (10 mM HEPES, 140 mM NaCl, 2.5 mM $CaCl_2$, pH 7.4) were added to the samples, which were incubated for 15 min at room temperature in the dark. After incubation, 470 μL of binding buffer was added, and the samples were immediately analysed by flow cytometry. To limit background noise from dust and crystals, all reagents were filtered twice with 0.22 μm filters. Flow cytometer calibration was performed using a mixture of fluorescent calibration beads (0.16, 0.20, 0.24, 0.5 μm, Megamix-Plus SSC, Biocytex, Marseille, France) specifically designed for EVs determination following the manufacturer's instructions (Figure 1(A,B)) [26]. EVs were identified as events inside the EVs region, positive for annexin V and immuno-reactive to cell-specific monoclonal antibody (Figure 1(C,D) and Supplemental Table S1). To quantify the absolute number of EVs, counting spheres (6 μm) with a known concentration (around 1000 spheres/μL, Perfect-Count Microspheres, Cytognos, Salamanca, Spain) were added to each sample as an internal standard. Samples were processed by flow cytometry for 2 min at a low flow rate for EVs determination, and the results were expressed as events/mL.

D-dimer, sP-selectin and antigenic TF measurement

D-dimer, sP-selectin and antigenic TF levels in PPP were measured in both VTD and cancer groups with

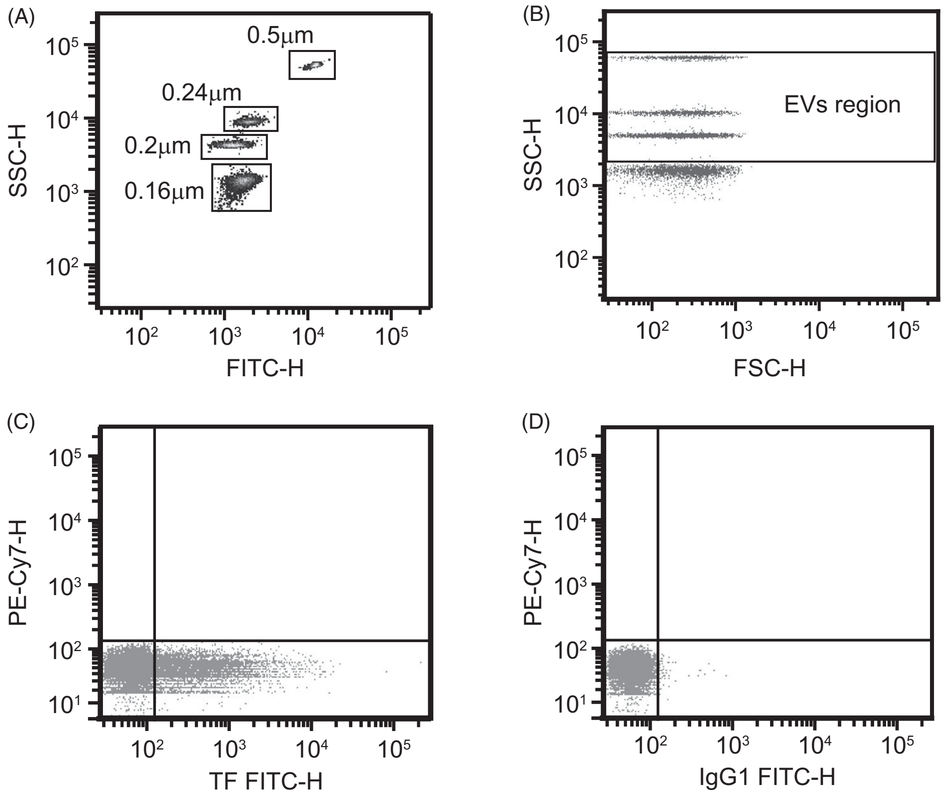

Figure 1. Extracellular vesicles identification by flow cytometry. (A, B) Calibration of flow cytometer using Megamix-Plus SSC beads to set up gate limits for EVs detection. (A) Identification of bead subpopulations based on their complexity properties and fluorescent intensity. (B) EVs gate establishment, 0.5 μm (upper cloud), 0.24 μm (second cloud), 0.20 μm (third cloud), 0.16 μm (bottom cloud). The defined EVs region (within two horizontal lines) covers most of the theoretical range of EVs size (0.1 μm to 1 μm). (C, D) Representatives dot plots illustrating the EVs subpopulation from a patient with venous thromboembolism disease. (C) EVs carrying TF identified in the bottom right region TF marker ($CD142^+$) and (D) its respective isotype-matched negative control antibody. IgG1: immunoglobulin G1; H: height signal.

commercial ELISA kits following the manufacturer's instructions (Acute Care™ D-dimer test pack, Siemens Healthcare Diagnostics, sensitivity: 6 μg/L, specificity: cross-linked fibrin degradation products, D-dimer, Newark, DE, USA; Human sP-selectin/CD62P Immunoassay, sensitivity: 0.121 ng/mL, specificity: recombinant and human P-selectin, R&D Systems, Minneapolis, MN, USA and Imubind® human Tissue Factor ELISA, sensitivity: 10 pg/mL approximately, specificity: TF-apo, TF and TF-VII complex and is designed such as there is no interference from other coagulation factors or inhibitors of procoagulant activity, American Diagnostica, Stamford, CT, USA). D-dimer, TF and sP-selectin levels were expressed as μg/L, pg/mL and ng/mL, respectively.

Statistical methods

The distribution of continuous variables was assessed by the Kolmogorov–Smirnov test. Categorical variables were expressed as frequencies or percentages and compared by $\chi 2$ statistics or Fisher's exact test. Continuous variables were summarized as median and 25th–75th percentiles and compared using Student's *t* test (for normal data) and Mann–Whitney *U* test in two groups and Kruskal–Wallis test in more than two groups (for non-normally distributed variables). For the cancer population data, they were considered outliers if they did not fit into the following intervals $Q1 - 1.5 \times IQR$; $Q3 + 1.5 \times IQR$. A *p*-value $<.05$ was considered statistically significant. All analyses were performed using IBM Products Statistics SPSS V.20.0 ©Copyright IBM Corp.1989, 2011.

Results

Demographic and clinical characteristics of the study population

The final study population consisted of two groups including 138 unprovoked VTD and 67 cancer patients, as shown in the patient enrolment flow chart in Figure 2. In this figure, we have named inconsistent

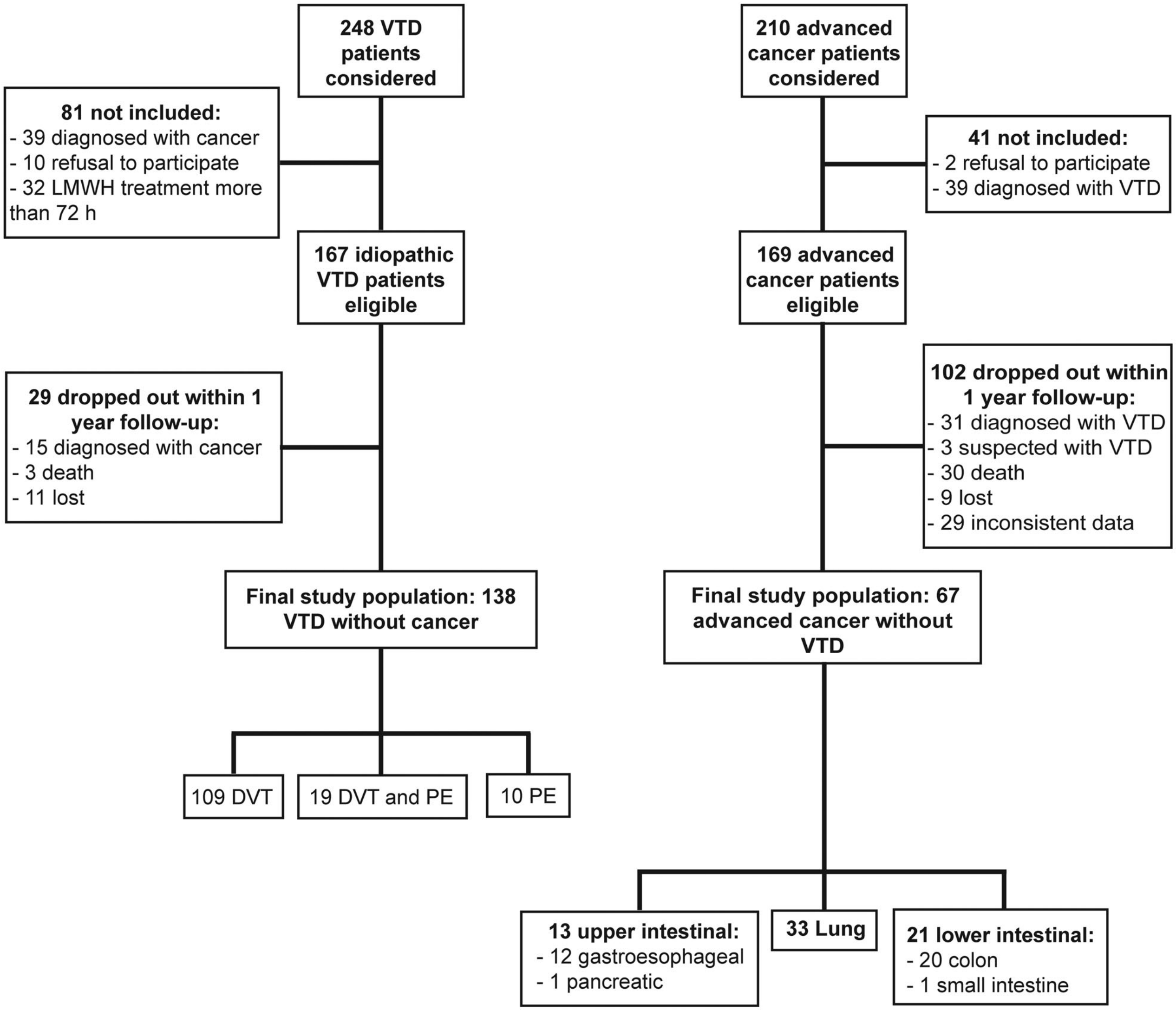

Figure 2. Flow chart of patient enrolment.

data under the following circumstances: patients with neoplastic disease other than stage IIIB or IV, patients without histological cancer confirmation, or patients in whom PE could not be ruled out as a cause of death.

Demographic and clinical characteristics were compared between groups (Table 1). The cancer group was characterized by a significantly higher prevalence of males, active smokers and regular alcohol consumers, in comparison to unprovoked VTD group. Instead, VTD group showed patients with a significantly higher prevalence of elevated body mass index (BMI), hypertension and arthrosis. All cancer patients were in an advanced stage, as stage IV tumour progression was found in 89.1% (57/67) of them. At the time of the enrolment, 73.1% (49/67) of cancer patients were receiving chemotherapy and 1.5% (1/67) had received chemotherapy and radiotherapy during the previous month. Among the VTD patients, 86 out of 138 (62.3%) had received anticoagulant treatment up to 72 h when the blood samples were obtained. Of these, 83.7% (72/86) of patients had been treated with therapeutic doses of enoxaparin.

Biomarkers profiling in VTD and cancer patients

Circulating EVs profiles for patients with unprovoked VTD and those with non-VTD associated cancer patients were compared in a bivariate analysis (Table 2). Numbers of total EVs, PEVs, LEVs and EEVs were similar in both groups (Table 2). PEVs were the most prevalent, representing nearly 80% of total EVs in both unprovoked VTD and non-VTD associated cancer patients. LEVs and EEVs were minority in both groups, representing only 1% of total EVs.

Regarding procoagulant EVs, a significant difference was observed between groups when TF^{+}EVs were

Table 1. Clinical and demographic characteristics of the cancer and VTD groups.

	VTD $n = 138$	Cancer $n = 67$
Age (years); median (P25–P75)	66.7 (53.7–76.7)	66.0 (58.0–72.0)
Male**	77 (55.8)	47 (70.1)
BMI (kg/m^2)^{a,*}	30.5 (27.8–34.6)	26.2 (23.3–28.5)
Smoker*	48 (34.8 %)	45 (67.2 %)
Alcohol consumption*	12 (8.7 %)	17 (25.4 %)
Hypertension*	72 (52.2 %)	22 (32.8 %)
Dyslipidemia	39 (28.3 %)	18 (26.9 %)
Diabetes	19 (13.8 %)	13 (19.4 %)
Chronic respiratory disease	31 (22.5 %)	13(19.4 %)
COPD	12 (8.7 %)	8 (11.9 %)
Asthma	8 (5.8 %)	1 (1.5 %)
Tuberculosis	2 (1.4 %)	2 (3.0 %)
Bronchiectasis	3 (2.2 %)	0 (0 %)
OSA	4 (2.9 %)	2 (3.0 %)
Peripheral artery disease	2 (1.4 %)	4 (4.5 %)
Dyspepsia	6 (4.3 %)	0 (0 %)
Intestinal inflammatory disease	3 (2.2 %)	1(1.5 %)
Liver disease	7 (5.1 %)	3 (4.5 %)
Ischaemic heart disease	9 (6.5 %)	2 (3.0 %)
Chronic heart failure	1 (0.7 %)	2 (3.0 %)
Arthritis	6 (4.3 %)	0 (0 %)
Arthrosis*	16 (11.6 %)	0 (0 %)
Chronic kidney disease	10 (7.2 %)	4 (6.0 %)
Neurological disorders	21 (15.2 %)	5 (7.5 %)

Quantitative variables are expressed as median (P25–P75). Qualitative variables are expressed as absolute values and percentages.
[a]Number of VTD = 121 and number of cancer = 59.
Significant p values are marked: $*p < .01$, $**p < .03$.
VTD: venous thromboembolism disease; BMI: body mass index; COPD: chronic obstructive pulmonary disease; OSA: obstructive sleep apnoea syndrome.

Table 2. Circulating extracellular vesicles, D-dimer, P-selectin and antigenic tissue factor levels in VTD and cancer.

	VTD Median (P_{25}–P_{75})	Cancer Median (P_{25}–P_{75})
Total EVs ($\times 10^3$)	13,489 (9037–22,558)	14,829 (10,854–20,319)
PSGL1$^+$EVs	4383 (3161–8582)	4900 (2293–7308)
TF$^+$EVs*	15,021 (7232–30,614)	7820 (4329–10,676)
EVs from endothelium		
EEVs	23,680 (15,588–32,913)	19,740 (16,093–30,516)
TF$^+$EEVs**	0.00 (0.00–530)	0.00 (0.00–340)
EVs from platelets		
PEVs ($\times 10^3$)	10,911 (5386–17,022)	11,578 (7729–17,055)
TF$^+$PEVs**	3390 (1297–10,714)	2400 (1252–4958)
EVs from leukocytes		
LEVs	85,460 (48,743–113,471)	70,960 (51,530–105,709)
PSGL1$^+$LEVs	1654 (704–2619)	1800 (881–2653)
D-dimer (µg/L)*	3890 (2959–8740)	764 (516–1598)
sP-selectin (ng/mL)*	50 (36–67)	36 (28–47)
Antigenic TF (pg/mL)*	240 (193–310)	203 (162–248)

Results are expressed as median (P25; P75).
Number of VTD are between 127 and 138; number of cancer are between 63 and 67.
EVs values are presented as number of events/mL unless specified otherwise.
VTD: venous thromboembolism disease; EVs: extracellular vesicles; PSGL1; P-selectin ligand 1; TF: tissue factor; EEVs: endothelial cell-derived EVs; PEVs: platelet-derived EVs; LEVs: leukocyte derived; sP-selectin: soluble P-selectin. Significant p values are marked: $*p < .01$, $**p = .02$.

analysed. A double amount of TF$^+$EVs was detected in patients with VTD compared to those with cancer (Figure 3(A)). When the cellular origin was analysed, significantly higher levels of platelet-derived (TF$^+$PEVs) and endothelial-derived (TF$^+$EEVs) were observed in patients with thrombosis compared to those with cancer (Table 2). Regarding PSGL1$^+$EVs, no difference was found neither in the total number of PSGL1$^+$EVs nor that with leukocyte origin (PSGL1$^+$LEVs) between VTD and cancer patients (Table 2).

Classical coagulation-related blood parameters were also evaluated. Plasmatic D-dimer concentration in VTD patients was fivefold higher than that in cancer ones. An increase in plasmatic sP-selectin and TF contents was also observed in VTD patients compared to cancer patients (Figure 3(B,D), Table 2). These observations indicated different plasmatic profiles of TF$^+$EVs, D-dimer, sP-selectin and antigenic TF between thrombosis and cancer patients.

Thereafter we evaluated whether the increase in TF$^+$EVs observed in VTD patients were really characteristic of the disease development. For that, a second analysis was performed comparing thrombosis and cancer patients with healthy subjects. In this analysis, we only considered variables that previously were significantly different between thrombosis and cancer groups. VTD patients had higher BMI and cancer patients were more frequently male, active smokers and regular alcohol consumers compared to healthy subjects. As expected, we observed an elevated level of TF$^+$EVs in VTD compared to healthy subjects ($p < .001$). TF$^+$PEVs and TF$^+$EEVs were also increased in VTD compared to healthy subjects, although not statistically significant ($p = .148$ and .229, respectively). On the other hand, although a trend of increase in TF$^+$EVs in the cancer group was observed, it did not reach a significant difference with respect to the healthy group ($p = .194$, Table 3). These observations indicated that elevated levels of TF$^+$EVs in VTD patients were specifically associated with their disease development in comparison to cancer patients and healthy subjects.

Comparative analysis between VTD patients and specific subgroups of cancer patients was also carried out to evaluate EVs contents and clinical-demographic characteristics in these subgroups. The cancer group was classified according to neoplasia localization in 3 subgroups: lung, upper intestinal and lower intestinal.

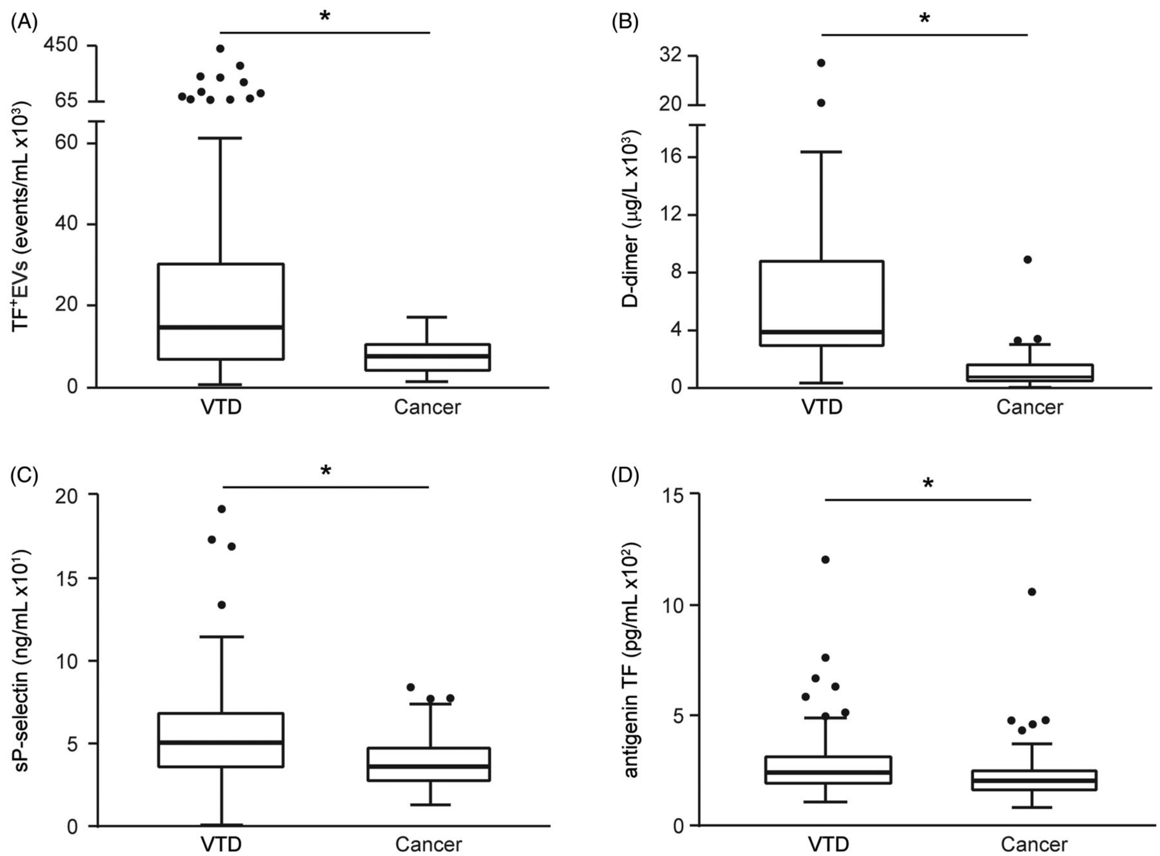

Figure 3. Plasma marker profiles in VTD and cancer. (A) TF^+EVs. (B) D-dimer. (C) sP-selectin. (D) antigenic TF. Results are expressed as median (horizontal line inside box), 75th and 25th percentiles (upper and lower horizontal lines delimit the box respectively) and 1.5-fold interquartile range (whiskers). Outliers/extreme values (circles); $^*p < .05$. Mann–Whitney U test was used for the comparison of all determinations.

Table 3. Comparison of clinical, demographic and circulating TF^+EVs features of VTD and cancer patients to healthy subjects.

	Healthy subjects $n = 14$	VTD $n = 138$	Cancer $n = 67$
Male	5 (35.7)	77 (55.8)	47 (70.1)**
BMI[a]	23.09 (5.40)	30.5 (6.9)*	26.2 (5.5)
Smoker	1 (7.1)	48 (34.8)	45 (67.2 %)***
Alcohol consumption	0 (0.0)	12 (8.7)	17 (25.4 %)**
Hypertension	0 (0.0)	72 (52.2)*	22 (32.8 %)***
Arthrosis	0 (0.0)	16 (11.6)	0 (0 %)
TF^+EVs	4243 (2638–7054)	15,021 (7232–30,614)*	7820 (4329–10,676)
TF^+ PEVs	2330 (1568–3420)	3390 (9470)3389 (1297–10,714)	2400 (1252–4958)
TF^+EEVs	0.00 (0.00–200)	0.00 (0.00–530)	0.00 (0.00–340)

Quantitative variables are expressed as median (P25–P75). Qualitative variables are expressed as absolute values and percentages. MPs values are presented as number of events/mL.
Significant p values are marked: $^*p < .01$ for VTD versus healthy subject; $^{**}p = .02$ and $^{***}p < .01$ for cancer versus healthy subjects.
[a]Number of VTD = 121 and number of cancers = 59; significant p values are markedly in bold. EVs values are presented as number of events/mL.
VTD: venous thromboembolism disease; BMI: body mass index; EVs: extracellular vesicles MPs: microparticles; TF: tissue factor; TF^+EVs: EVs carrying TF; TF^+PEVs: platelet-derived EVs carrying TF; TF^+EEVs: endothelial cell-derived EVs carrying TF.

We observed that TF^+EVs levels were similar among cancer subtypes and were maintained significantly higher in VTD compared to each cancer subtype (Figure 4). The complete results are shown in Supplemental Tables S2 and S3. TF^+EVs, among all EVs determinations, were still the only determination that significantly differed between VTD and the rest of groups: cancer, healthy subjects, lung cancer, upper intestinal cancer and lower intestinal cancer. These observations support that TF^+EVs differed specifically between unprovoked VTD and advance stage cancer patients independent of the cancer subtype.

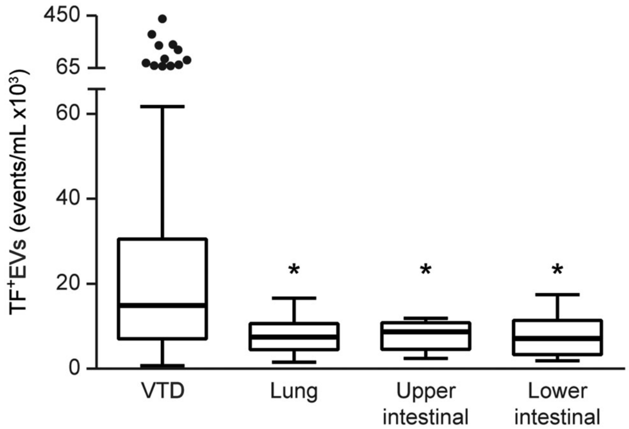

Figure 4. EVs carrying tissue factor in VTD and subtypes of cancer. Results are expressed as median (horizontal line inside box), 1.5-fold interquartile range (whiskers), 75th and 25th percentiles (upper and lower horizontal lines delimit the box respectively). Outliers/extreme values (circles); $^{*}p < .05$ compared to VTD group. Kruskal–Wallis test was used comparing four groups.

Discussion

The development of VTD in cancer patients is associated with a significantly worse prognosis. In particular, VTD seems to increase mortality rate especially in patients with advanced stages of the disease [27]. This led to the need to stratify cancer patients according to their risk to develop VTD [22], which is a clinical challenge, and to the search for biomarkers, which are essential for early diagnosis and management of VTD in these patients. In the current study, we demonstrated an elevated TF$^+$EVs content in patients with unprovoked VTD compared to patients with an advanced stage of cancer without the development of thromboembolism. The potential role of TF$^+$EVs as biomarkers of thrombosis development in cancer patients has been investigated using *in vivo* and *in vitro* experimental models [20,28–30]. In human studies, a case–control study comparing cancer patients with and without acute VTD demonstrated an association between elevated TF$^+$EVs and VTD-associated cancer [31]. Similar results were found by other groups [32,33]. However inconsistent retrospective observations have been published in which multivariate analysis failed to show a significant association between elevated TF$^+$EVs and VTD in cancer patients [34]. In addition, results from prospective studies are also inconclusive [35–38]. Pre-analytical and analytical conditions, study design and selection bias could explain these controversies. Moreover, it is especially difficult to clarify biomarkers, including TF$^+$EVs, which are related to the inflammation or invasive neoplastic process in addition to a procoagulant status. To circumvent this difficulty, we prospectively selected VTD patients free from cancer and cancer patients free from VTD. Our results showed marked difference in TF$^+$EVs between both patient groups. TF$^+$EVs were significantly higher in VTD patients than advanced cancer patients in total EVs and in EVs from specific cellular origin (endothelium and platelet). This observation is independent of the cancer subtype studied here as similar results were obtained when comparing TF$^+$EVs levels in VTD to each subtype (lung, low or high gastro-intestinal cancer). No statistical difference in TF + EMPs and TF + PMPs was observed when comparing VTD group to each subgroup of cancer studied here (lung cancer, upper intestinal cancer, lower intestinal cancer, see supplementary data). It could be due to the effect of sample size on statistical power.

As expected, TF$^+$EVs levels in healthy subjects were lower than the VTD and cancer patients, although only the difference between VTD patients and the healthy subjects was significant. This supports TF$^+$EVs being characteristic of VTD events. The increase in TF$^+$PEVs and TF$^+$EEVs in VTD compared to healthy subjects did not reach statistical significance. This could be the result of the high variability among samples and the low sample size of the heathy subject group.

The finding that TF$^+$EVs characterize the venous thromboembolism entities more than different types of cancer opens, according to our opinion, new research fields. Future studies are necessary to evaluate whether TF$^+$EVs could serve as a diagnostic or prognostic biomarker of VTD in cancer patients. In addition to EVs, we also measured plasma markers of active thrombotic processes and found that D-dimer, sP-selectin and TF antigen are markedly elevated in VTD compared to cancer patients. Clinical trials and prospective studies have suggested D-dimer with a strong diagnostic value in VTD [39–41]. Indeed, D-dimer measurement is included in algorithms that are commonly employed to diagnose VTD [42]. However, the diagnostic value of D-dimer in the algorithms of VTD in patients with cancer remains under debate. Although D-dimer is strongly associated with VTD, it can be increased in pro-inflammatory and neoplastic settings. In particular, it has been questioned whether the cut-off level of 500 μg/L used in VTD diagnostic algorithms is appropriate in specific populations such as cancer patients, who have higher D-dimer values compared to the general population [43]. In our cancer group, the median (IQR) for D-dimer was 764 (1454) μg/L suggesting that a cut-off of 1000 ug/L might be more suitable in a VTD diagnostic algorithms for patients with cancer.

Elevated levels of sP-selectin were found in thrombosis compared to advanced stage cancer patients. This leads to the proposition that, despite of an increase in sP-selectin in patients with metastatic cancer, their values rise in acute VTD. A previous meta-analysis report including 11 studies concluded sP-selectin may have a potential role in the diagnosis of VTD [44]. Other authors found sP-selectin may have a predictive role in the diagnosis of VTD in patients with cancer demonstrating that when both D-dimer and sP-selectin are included in the Khorana predictive scale, the accuracy improves [45]. These authors recommend cut-offs for D-dimer $\geq$ 1440 μg/L and sP-selectin $\geq$ 53.1 ng/mL as potential values in predicting thrombotic events in patients with cancer. These proposed cut-offs are in accordance with the current study in which values for cancer patients without VTD are lower than these concentrations.

Antigen TF has also been demonstrated to be a potential biomarker of recurrent VTD in patients with cancer who are on anticoagulation treatment [46]. Our results of high TF values in VTD patients are consistent with the CATCH trial with 900 patients, in which the highest quartile of TF experienced is associated with the greatest VTD recurrence. However, the achieved results with this biomarker, despite reaching a significant difference in the comparison of medians, may not have clinical implication, as the overlap between the P25–P75 values is evident (Figure 2(D)).

Individually, TF^{+}EVs, D-dimer, sP-selectin and even antigen TF, all have been reported the ability to identify cancer patients at risk of venous thromboembolism. It remains to be studied the usefulness of using all these biomarkers together in clinical settings. Future studies are necessary to determine whether increased sensitivity obtained with the addition of the D-dimer and P-selectin on the scale of Khorana [45,46] could be further improved with the addition of TF^{+}EVs.

In addition to blood samples used in the current study, it could be interesting to determine TF^{+}EVs in tumour biopsies from cancer patients for its potential usefulness and applicability to clinical practice, which is completely unknown.

In summary, in the current study, it was demonstrated that circulating EVs, specifically TF^{+}EVs, in combination with plasmatic markers of hypercoagulable states, such as D-dimer, sP-selectin and may be antigen TF, are able to discriminate between cancer patients without venous thromboembolism and patients with unprovoked VTD. It would be interesting to investigate in future studies whether these biomarkers together serve as predicting thrombotic events in cancer populations, as well as other possible applications of the EVs determination.

Authors' contributions

VSL designed research, performed research, interpreted data, and wrote the manuscript; LG performed research, interpreted data, and wrote the manuscript; MFG designed research, analysed and interpreted data, and critically reviewed the manuscript; EAO analysed and interpreted data, and critically reviewed the manuscript; TEH and LJP recruited patients, analysed and interpreted data, and critically reviewed the manuscript; MJC and FJRM analysed and interpreted data, and critically reviewed the manuscript; JLLB, ABQ and VVL recruit patients, analysed and interpreted data. ROC coordinated the study, designed research and critically reviewed the manuscript. All authors gave their consent to the final version of the manuscript.

Disclosure statement

Authors do not have any conflicts of interest to declare. All authors have read the Annals of Medicinés policy on disclosure of potential conflicts of interest. All authors have read the journal's authorship agreement, and the article has been reviewed and approved by all named authors.

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